

STAT 362 Final Project

Spring 2026

Due: Tuesday, June 9

1 Introduction

For the final project for STAT 362, you will pick one of the two following options: **Theory** or **Multimedia Application**. You can find the details for both options below:

1.1 Theory

If you choose the Theory option, you are required to find **a Machine Learning (ML) model that is not covered in STAT 301/303 or STAT 362** and work on the theory behind it. You are allowed to choose an extension of a model that is covered in STAT 301/303 or STAT 362.

Your model may be a supervised learning model, unsupervised learning model, or something that falls under another ML topic. Whichever you choose, you are required to understand and explain (1) how the model is formulated and (2) how its training algorithm works **by going through the derivations and any other related mathematical tools**. You are also required to explain your math in technical terms both in a report and in a recorded presentation.

Lastly, you are required to find a benchmark dataset to demonstrate the important aspects of the model you choose. There are no minimum requirements (on the number of rows/columns, etc.) for the dataset, since the main focus of this option is theory. The dataset can be synthetic or real-life. As long as you demonstrate the aspects that make your model significant for ML clearly in your results, you are allowed to use any dataset.

To apply your model to the benchmark dataset, you may either implement the model yourself or use an already implemented version from any Python library.

If you choose an extension of a model that is covered in STAT 301/303 or STAT 362, you are also required to compare your model to its original version, both mathematically and on the benchmark dataset.

1.2 Multimedia Application

If you choose the Multimedia Application option, you are required to find or put together a **real-life dataset that includes more than one data modality**. Some examples for different data modalities are (but are not limited to) structured (tabular) format, images, videos, text, and time-series data, such as sound recordings and signals. Your dataset must include **at least two different modalities**.

Please note that you must choose or put together a **real-life** dataset, so you are not allowed to create data yourself or put together synthetic data from different sources. You will need to cite your data sources.

There is no minimum requirement for data size; however, if your results indicate a low performance with the lack of enough data as an obvious reason, you will get half credit at best from your project.

To process your multimedia dataset, your ML workflow is required to include **models that are specialized on specific formats**, such as Convolutional Neural Networks for image data, Recurrent Neural Networks for time-series data, or classical ML models for tabular data.

You are allowed to use the models that are covered in STAT 301/303 and STAT 362, and also any model that is not covered in any of these courses. Your models can be supervised, unsupervised, or something that falls under another ML topic. Whichever you choose, you are required to understand and explain (1) how each model in your workflow functions and (2) why it can handle data in a specific format better than other models. You may or may not use any math to explain your workflow, but you must have **a profound understanding of how each ML component functions in your workflow**. You are also required to explain all parts of your workflow in technical terms both in a report and in a recorded presentation.

After you run your ML workflow on the multimedia data, you are required to (1) display your results as well as any other numeric/visual output that is relevant and (2) interpret your results, being mindful of the ML task (supervised, unsupervised, etc) and the modality of the data at hand.

As long as you present and interpret your results well, you will not be penalized for low ML performance. As also written above, the only exception for this is the lack of sufficient data as a reason for the low ML performance.

2 Deliverables:

- **The proposal:** By the end of Week 7, you are required to submit a project proposal that will state (1) the option you choose, (2) the model (for Theory) or dataset(s) (for Multimedia Application) you are planning to work on and (3) an initial plan for your project.

You are allowed to change your topic after the proposal deadline **only with instructor**

approval.

Also note that the instructor must approve your submitted proposal as a valid project. Therefore, **submitting a proposal is mandatory**; any project without a submitted and approved proposal will not receive any credit.

- **The report:** You are required to submit a written report that (1) introduces and explains the content and the scope of your project, (2) includes the body of your work (the math and derivations for Theory or the dataset and ML workflow for Multimedia Application), (3) displays and interprets your results, and (4) cites your sources.

There are no minimum/maximum length requirements for your report. Your content in the report will be graded on its (1) factual and technical correctness, (2) consistency, (3) completeness, and (4) clarity.

A template for the report will be provided by the end of Week 7.

- **The dataset:** You are required to submit your dataset(s) along with the write-up. You can submit the dataset(s) as-is (which is viable only for Theory) or as a zip file (for either option).

If your data volume is too large to submit on Canvas, please contact your instructor in a timely manner.

- **The presentation:** Along with the report and the dataset(s), you are required to submit a 5-10-minute recorded presentation to explain your work in a user-friendly way.

For the presentation recording, there will not be any template that you need to follow. Any software and format is acceptable to record (and submit) your presentation.

A few strict requirements for the presentation are: (1) All relevant content in your report must be at least touched upon or discussed in detail; (2) both your face and your presentation screen must be visible in the recording; and (3) you must demonstrate a good understanding of your written content through your presentation skills, i.e. you may not simply copy sentences and figures from your report and read off slides.

Please note that for a project to be complete, you must submit the report, the dataset and the presentation. Any incomplete project will not receive any credit.